

Effects of dietary protein on composition and metabolism of plasma lipoproteins in rabbits.



Changes in the concentration and composition of serum VLDL, LDL, and HDL were studied in rabbits transferred from Chow diets to cholesterol-free, semipurified diets containing casein or isolated soy protein. During the first week on the casein diet, there was a marked increase in LDL-cholesterol and these higher levels were maintained during the subsequent 3 weeks of the study. Similar but less marked changes were obtained with the soy protein diet. When the percent composition of the particles was determined, both VLDL and LDL had a higher proportion of cholesterol. Turnover studies indicated that the FCRs for radiolabelled VLDL and LDL were reduced in casein-fed animals compared to those fed soy protein. The elevated LDL levels in casein-fed rabbits were primarily due to a reduction in receptor-mediated catabolism of LDL-apo B. Receptor-independent removal in the two groups was similar. These studies show that the hypercholesterolemia in casein-fed rabbits, compared to those fed soy protein, is associated with cholesterol enrichment of LDL and impaired receptor-dependent removal of LDL-apo B.